Predictive Warm Pools for Serverless Cold-Start Reduction

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Abstract

Serverless platforms start a fresh sandbox whenever a request arrives for a function that has no idle instance, and this cold start can add hundreds of milliseconds to the response time. Platforms therefore keep finished instances warm for a fixed period, which wastes memory on functions that are rarely called and still misses bursts on functions that are called often. We present WarmCast, a controller that forecasts the arrival rate of each function and sizes a pool of pre-started instances to minimise the sum of memory cost and cold-start cost. On a two-week production trace, WarmCast reduces the cold-start rate from 18.4% to 6.1% while using 22% less idle memory than a ten-minute keep-alive policy.

1 Introduction

Function-as-a-Service (FaaS) lets developers deploy individual functions and pay only for the time they run [7]. The platform hides servers, scaling and placement, but it cannot hide the cost of starting a sandbox. When a request finds no idle instance of its function, the platform must create a container or microVM, load the language runtime and initialise the application, which together take between 100 ms and several seconds [10, 8]. Such cold starts dominate the tail latency of short functions.

Most platforms use a static keep-alive policy: after an instance finishes a request, it stays in memory for a fixed period, typically ten minutes, in case another request arrives [9]. This policy is easy to deploy, but it treats every function the same. A function invoked once per hour holds memory for ten minutes and is still cold on its next call, while a function with bursty traffic needs several instances at once and receives only one.

We observe that the arrival rate of most functions changes slowly compared with the time needed to start an instance. A short-term forecast of the rate is therefore enough to decide how many instances to keep ready. Building on this observation, we make three contributions. First, we model the number of arrivals in the next interval as a Poisson variable whose rate is estimated online (Section 3). Second, we derive the pool size that minimises the expected cost of idle memory and cold starts, which has a closed form for a given rate. Third, we evaluate the resulting controller, WarmCast, on a production trace and show that it dominates fixed keep-alive policies on both cost and latency (Section 4).

2 Related Work

Prior work reduces cold starts in two ways: by making each start cheaper, or by starting fewer instances. On the first front, SOCK [8] caches imported packages in a zygote process, Firecracker [1] boots a minimal microVM in about 125 ms, and Catalyzer [3] restores an initialised instance from a checkpoint. SAND [2] runs functions of the same application in one container to share their start cost. These techniques shorten each cold start but do not change how often one occurs.

On the second front, Shahrad et al. [9] characterise the Azure Functions workload and propose a hybrid histogram policy that chooses the keep-alive and pre-warming windows for each application. FaasCache [4] treats warm instances as a cache and evicts them with a Greedy-Dual policy. Both approaches reason about the lifetime of one instance and leave the number of concurrent instances to the autoscaler. WarmCast instead decides how many instances to keep, which matters for functions whose requests arrive in bursts.

Our forecaster is deliberately lightweight. We use exponential smoothing [6] because it needs constant memory per function and adapts within a few intervals; heavier models such as recurrent networks would

cost more to run than the cold starts they avoid for most functions. Platforms built on OpenLambda [5] expose the per-minute counts that the forecaster needs.

3 Method

WarmCast runs as a controller next to the platform scheduler. Time is divided into intervals of length τ, which we set to one minute. At the start of each interval the controller updates a rate estimate for every function, computes a target pool size, and starts or stops pre-warmed instances to reach it. Requests are routed to warm instances first; a request that finds none triggers a cold start as before.

Rate estimation. Let x_t be the number of requests a function received in interval t. We estimate its arrival rate with exponential smoothing,

$$\hat{\lambda}_t = \alpha \, \frac{x_t}{\tau} + (1 - \alpha) \, \hat{\lambda}_{t-1}, \tag{1}$$

where $\alpha \in (0, 1]$ controls how quickly the estimate follows a change in load. A large α reacts to bursts but also to noise; we use $\alpha = 0.3$, which performed best on a held-out day of the trace.

Pool sizing. Given the rate, we model the number of arrivals N in the next interval as a Poisson variable with mean $\mu = \hat{\lambda}_t \tau$. If k instances are warm, every arrival beyond the k-th concurrent one causes a cold start. Let c_m be the cost of keeping one instance in memory for an interval and c_s the cost of one cold start. The expected cost of a pool of size k is

$$C(k) = c_m \tau \, k + c_s \, \mathbb{E}\big[(N - k)^+\big], \tag{2}$$

and the controller chooses $k^* = \arg\min_{k \ge 0} C(k)$. Because C is convex in k, the minimiser is the smallest k for which $\Pr[N > k] \le c_m / c_s$, which we find by evaluating the Poisson tail from $k = 0$ upward.

Cost parameters. We express both costs in GB-seconds of memory. The memory cost c_m is the memory size of the function times τ, and the start cost c_s is the memory size times the measured cold-start duration, multiplied by a weight w that encodes how much the operator values latency over memory. With $w = 1$ the controller minimises total memory time; larger values buy lower latency with more memory.

Safety limits. To avoid holding memory for a function whose traffic has stopped, the controller caps k^* at the largest number of concurrent requests observed in the last hour and releases all instances of a function that received no request for 30 minutes. Pre-warming uses spare capacity only; when a host runs short of memory, pre-warmed instances are evicted before instances that serve requests.

4 Evaluation

We replay the first two weeks of the Azure Functions trace [9], which records per-minute invocation counts, execution times and memory sizes for 46 413 functions. We use day 1 to tune α and w and report results on days 2–14. Cold-start durations are drawn from the per-runtime distribution measured by Wang et al. [10].

We compare WarmCast with three policies: a fixed keep-alive of 10 minutes (*KA-10*), the same policy with a 60-minute window (*KA-60*), and the hybrid histogram policy of Shahrad et al. (*Hybrid*). All policies run on the same simulated cluster of 200 hosts with 64 GB of memory each.

Table 1 summarises the results. WarmCast lowers the cold-start rate to 6.1%, compared with 18.4% for KA-10 and 9.7% for Hybrid, while holding less idle memory than either. KA-60 reaches a similar cold-start rate only by keeping almost four times as much memory idle. The 99th-percentile latency follows the cold-start rate, since most requests in the tail are cold.

Figure 1 breaks the cold-start rate down by how often a function is invoked. Rarely invoked functions are cold under every policy, because no forecast can predict a single request per day. The gain of WarmCast

Table 1: Results on days 2–14 of the trace. Idle memory is the average memory held by idle instances.

Policy	Cold starts (%)	P99 latency (ms)	Idle memory (TB)
KA-10	18.4	1240	2.31
KA-60	6.8	410	7.02
Hybrid	9.7	671	2.05
WarmCast	6.1	388	1.80

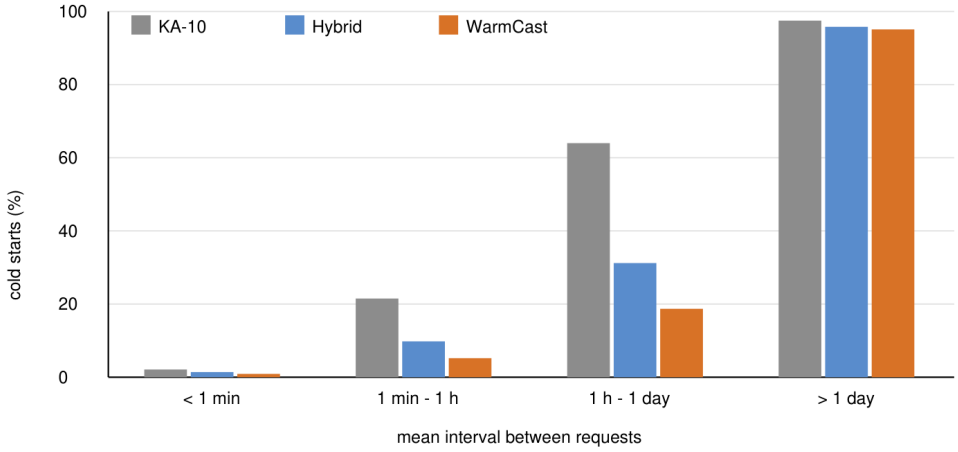


Figure 1: Cold-start rate by invocation frequency. Each group of bars covers the functions whose mean interval between requests falls in the given range.

comes from functions invoked between once per minute and once per hour, where fixed windows either expire too early or hold instances for too long.

The controller adds little overhead. Updating the estimate in Eq. (1) and searching for k^* take 3.2 μs per function on one core, so a single controller thread handles all 46 413 functions within 0.2 s per interval. We did not observe any measurable effect of the controller on request latency in a prototype deployment.

Our simulator is event-driven and models the memory of each host but not its CPU. It assumes that a warm instance serves a request immediately and that a cold start blocks the request for the full start duration. These assumptions favour no policy in particular, since all policies share them.

5 Conclusion

We presented WarmCast, which sizes a pool of warm instances for each serverless function from a short-term forecast of its arrival rate. Because the controller needs no changes to the functions themselves, it can be deployed on existing platforms without involving developers. The pool size follows from a convex cost model with a closed-form minimiser, and the controller runs in microseconds per function. On a production trace it reduces cold starts by a factor of three compared with the default keep-alive policy while holding less idle memory. In future work we plan to share warm instances between functions with the same runtime and to forecast with seasonal models.

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